Test Series No.

1.12

MASTERY TEST SERIES

DURATION: 180 Minutes | **M. MARKS: 720**

DATE: 19/03/2026

SYLLABUS

Physics:	Full Syllabus * As per NTA
Chemistry:	Full Syllabus * As per NTA
Botany:	Full Syllabus * As per NTA
Zoology:	Full Syllabus * As per NTA

GENERAL INSTRUCTIONS:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **3 hours** duration and the Test Booklet contains **180** multiple choice questions (four options with a single correct answer).
3. There is only one correct response for each question.
4. Each **correct** answer will give **4 Mark** while **1 Mark** will be **deducted** for a **wrong** answer. The maximum marks are 720.
5. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
6. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.

OMR INSTRUCTIONS:

1. Use only blue/black dark ballpoint pen.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Name of the Student (In CAPITALS): _____

Roll Number: _____

OMR Bar Code Number: _____

Student's Signature: _____ **Invigilator's Signature:** _____

1. A uniform flexible chain of mass M and total length L hangs partially from the edge of a rough horizontal table. The coefficient of static friction between the chain and the table is $\mu = 0.25$. What is the maximum fraction of the total length of the chain that can hang over the edge without the chain sliding down?

(1) 20% (2) 25%

(3) 33.3% (4) 50%

2. A solid copper sphere of radius R and a solid copper cube of side length a have the exact same volume. Both are heated to the same initial temperature T and are allowed to cool in identical surrounding environment. If R_s and R_c are the initial rates of cooling (rate of fall of temperature) of the sphere and the cube respectively, what is the ratio $R_s : R_c$?

(1) 1:1 (2) $\left(\frac{\pi}{6}\right)^{1/3}:1$

(3) $1:\left(\frac{6}{\pi}\right)^{1/3}$ (4) $2:\left(\frac{6}{\pi}\right)$

3. A Zener diode with a breakdown voltage $V_2 = 10$ V has a maximum power dissipation rating of 0.5 W. It is intended to be used in a simple voltage regulator circuit connected to an unregulated DC power supply of 20 V. What is the minimum value of the series resistance R_s that must be connected to ensure the Zener diode is not damaged when the load is completely disconnected?

(1) 100 Ω

(2) 400 Ω

(3) 50 Ω

(4) 200 Ω

4. One mole of an ideal monatomic gas undergoes an expansion process represented by $P \propto V$. If heat ΔQ is supplied to the gas during this process, what fraction of this heat gets converted into mechanical work?

(1) $\frac{1}{5}$ (2) $\frac{2}{5}$

(3) $\frac{1}{4}$ (4) $\frac{1}{3}$

5. A cell of internal resistance r and EMF E is connected to a variable external load R. It is observed that the power dissipated in the load remains the same whether R is adjusted to 2Ω or 8Ω. Furthermore, it is found that the maximum possible power that can be dissipated in this load is 9 W. Find the EMF E of the cell.

(1) 12 V

(2) 6 V

(3) 18 V

(4) 24 V

6. A particle of mass m starting from rest moves in a straight line under the action of a force that delivers a constant power P. What is the ratio of the distance covered by the particle in the first t_0 seconds to the distance covered in the time interval from $t = t_0$ to $t = 4t_0$?

(1) 1:8 (2) 1:9

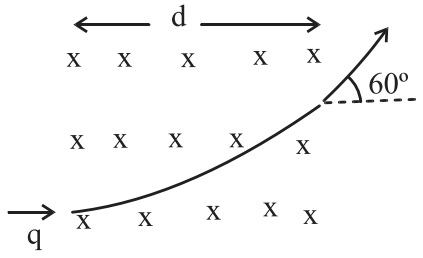
(3) 1:4 (4) 1:7

7. A uniform thick rope of total length L, cross-sectional area A, total mass M, and Young's modulus Y is suspended vertically from a rigid ceiling. Under its own weight, the rope undergoes a slight elongation. What is the total elastic potential energy stored in the rope due to its own weight? (Assume g is the acceleration due to gravity)

(1) $\frac{M^2g^2L}{2AY}$ (2) $\frac{M^2g^2L}{4AY}$

(3) $\frac{M^2g^2L}{6AY}$ (4) $\frac{M^2g^2L}{8AY}$

8. A positively charged particle with mass m and charge q enters perpendicularly into a uniform transverse magnetic field B of width d. If it emerges out of the field region with its path deviated by 60º, the time spent in the field is:



(1) $\frac{\pi m}{qB}$ (2) $\frac{\pi m}{2qB}$

(3) $\frac{\pi m}{3qB}$ (4) $\frac{\pi m}{6qB}$

[Contd...

9. A thermally insulated rigid container holds a completely unreactive mixture of 2 moles of Helium gas and 3 moles of Oxygen gas (assuming no vibrational modes are active). What is the ratio of the specific heat at constant pressure to the specific heat at constant volume ($\gamma = C_P/C_V$) for this gaseous mixture?

(1) 1.48 (2) 1.50
(3) 1.54 (4) 1.67

10. A satellite of mass m is revolving in a stable circular orbit at a height 2R above the surface of the Earth (where R is the radius of the Earth and M is the mass of the Earth). What is the minimum extra kinetic energy that must be supplied to the satellite for it to permanently escape the Earth's gravitational pull?

(1) $\frac{GMm}{2R}$ (2) $\frac{GMm}{3R}$
(3) $\frac{GMm}{4R}$ (4) $\frac{GMm}{6R}$

11. A monochromatic light ray passes symmetrically through an equilateral glass prism, meaning the angle of incidence is exactly equal to the angle of emergence. If the angle of minimum deviation observed is equal to the refracting angle of the prism, what is the refractive index of the prism material?

(1) $\sqrt{2}$ (2) $\sqrt{3}$
(3) 1.5 (4) 2.0

12. A projectile is thrown from the origin on level ground with an initial velocity vector $\vec{u} = \left(3\hat{i} + 4\hat{j}\right)$ m/s. Assuming g = 10 m/s^2 and that the $\hat{j}$ unit vector points vertically upwards, at what time t will its instantaneous velocity vector be strictly perpendicular to its initial velocity vector?

(1) 0.40 s (2) 0.50 s
(3) 0.625 s (4) 0.80 s

13. Two flat, concentric circular coils have radii r and R, where r << R. If the planes of the two coils are mutually perpendicular, what is the mutual inductance M of this arrangement?

(1) $\frac{\mu_0 \pi r^2}{2R}$ (2) $\frac{\mu_0 \pi R^2}{2r}$
(3) Zero (4) $\frac{\mu_0 \pi r^2}{R}$

14. A small heavy bob of mass m is suspended from a fixed point by a massless, inextensible string of length L. At its lowest point, it is given a sharp horizontal kick, imparting an initial horizontal velocity $u = \sqrt{4gL}$. What is the tension in the string at the exact moment the string becomes perfectly horizontal? (g is acceleration due to gravity)

(1) mg (2) $2mg$
(3) $3mg$ (4) $4mg$

15. A capacitor of capacitance C is fully charged to a potential V and then cleanly disconnected from the battery. It is connected in parallel to another identical but completely uncharged capacitor. What is the ratio of the final total electrostatic energy stored in the system to the initial total electrostatic energy?

(1) 1:2 (2) 1:4
(3) 1: $\sqrt{2}$ (4) 3:4

16. In a series LCR circuit connected to a variable frequency AC source, the rms voltage across the inductor L, the capacitor C, and the resistor R are measured to be $V_L = 40$ V, $V_C = 40$ V, and $V_R = 20$ V respectively. What is the RMS voltage of the AC source and the power factor of the circuit at this instant?

(1) 100 V, 0.2 (2) 20 V, 1.0
(3) 20 V, 0.5 (4) 60 V, 1.0

17. A hollow tube is dipped vertically into water, creating an air column of length L closed at one end. This column resonates in its first overtone at 450 Hz. If the tube is lifted out of the water to become an open pipe of length L, what will be its fundamental frequency?

(1) 150 Hz (2) 300 Hz
(3) 450 Hz (4) 600 Hz

18. In an experiment to measure the acceleration due to gravity g using a simple pendulum, a student measures the length of the pendulum L as (100 ± 0.1) cm and the time period T as (2 ± 0.01) s. The maximum percentage error in the measurement of g is approximately:

(1) 0.1% (2) 0.5%
(3) 1.1% (4) 2.1%

[Contd...

19. A compound microscope consists of an objective lens of focal length f_0 = 2.0 cm and an eyepiece of focal length f_c = 5.0 cm An object is placed at a distance of 2.5 cm in front of the objective lens. If the final image is formed at the least distance of distinct vision (D = 25 cm), what is the magnitude of the total magnification produced by the microscope?

(1) 24 (2) 20

(3) 12 (4) 30

20. An electron is projected with an initial velocity $\vec{v} = v_0\hat{i}$ into a region of uniform magnetic field $\vec{B} = B_0\hat{j}$. While the electron moves in the field, it experiences a magnetic force. Regarding the de-Broglie wavelength (λ) associated with the electron during its motion, which of the following statements is correct?

(1) λ decreases continuously because the electron accelerates due to the magnetic force.

(2) λ varies periodically because the direction of the velocity vector changes continuously.

(3) λ remains constant because the magnetic force performs no work on the electron.

(4) λ increases because the magnetic field reduces the kinetic energy of the electron.

21. According to the Kinetic Theory of Gases, which assumption explains why the total energy of an isolated gas remains constant despite billions of molecular collisions occurring every second?

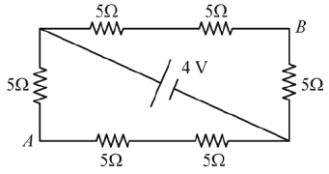
(1) The time of collision is negligible compared to the time between collisions.

(2) The collisions are perfectly elastic

(3) Gravity acts equally on all molecules

(4) Molecules move in straight lines between collisions.

22. In the given circuit, The potential difference between point A and B will be:



(1) $\frac{2}{3}$ V (2) $\frac{8}{9}$ V

(3) $\frac{4}{3}$ V (4) 2V

23. Given below are two statements: One is labelled as Assertion A and the other is labelled as Reason R:

Assertion (A): Work done by electrostatic forces in moving a charge between any two points in a uniform electric field is independent of the path followed by the charge between these points.

Reason (R): Electrostatic forces are non-conservative.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) A is true but R is false.

(2) A is false but R is true.

(3) Both A and R are true and R is the correct explanation of A.

(4) Both A and R are true and R is NOT the correct explanation of A.

24. Charge -1×10^{-6} C is on a drop of water having mass 10^{-6} kg. What uniform electric field should be applied on the drop so that it is in balanced condition with its weight? (g = 10 m/s^2)

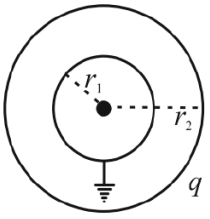
(1) 10 V/m vertically upward

(2) 10 V/m vertically downward

(3) 0.1 V/m vertically downward

(4) 0.1 V/m vertically upward

25. Two hollow concentric conducting spheres are of radii r_1 and r_2. The outer sphere is given a charge q and inner sphere is neutral. What is the charge developed on the inner sphere if it is grounded as shown?



(1) q (2) $-q$

(3) $-q\frac{r_1}{r_2}$ (4) $+q\frac{r_1}{r_2}$

26. The potential energy of a particle executing SHM is 2.5 J when its displacement is half of amplitude. The total energy of the particle is:

(1) 2.5 J (2) 5 J

(3) 12 J (4) 10 J

[Contd...

27. Two small drops of mercury, each of radius R coalesce to form a single large drop. The ratio of the total surface energy before and after the change is:
(1) $1:2^{1/3}$ (2) $2^{1/3}:1$
(3) 2:1 (4) 1:2

28. A solid cylinder of mass 2 kg and radius 4 cm is rotating about its axis at the rate of 3 rpm. The torque required to stop it after 2π revolutions is:
(1) 2×10^{-6} Nm (2) 2×10^{-3} Nm
(3) 12×10^{-4} Nm (4) 2×10^{6} Nm

29. A block of mass m is placed on a smooth wedge of inclination θ. The whole system is accelerated horizontally so that block does not slip on the wedge. The force exerted by the wedge on the block (N) is:
(1) $mg\cos\theta$ (2) $mg\sin\theta$
(3) $mg/\cos\theta$ (4) $mg/\sin\theta$

30. A circular coil of radius R carries a current I. The magnetic field at its center is B. The distance from the center on the axis where the magnetic field is B/8 is:
(1) $\sqrt{2}R$ (2) $2R$
(3) $\sqrt{3}R$ (4) $3R$

31. The ratio of the longest wavelength lines in the Balmer and Lyman series on the hydrogen spectrum is:
(1) 5:27 (2) 27:5
(3) 5:9 (4) 4:1

32. The magnetic susceptibility (χ) is negative for:
(1) Ferromagnetic material
(2) Superconductor
(3) Diamagnetic material
(4) Both (2) and (3)

33. An inductance L and a resistance R are first connected to a DC battery in series. After attaining steady state, the battery is disconnected but L and R remain connected in a closed circuit. Then, the current reduces to 37% of its steady state value in time:
(1) RL (2) $\frac{R}{L}$
(3) $\frac{L}{R}$ (4) $\frac{1}{LR}$

34. Various electromagnetic waves are given in **List-I** and various frequency values in **List-II**-Match the two Lists.

List-I		List-II	
(A)	Radio waves	(I)	1×10^{17} Hz
(B)	γ-rays	(II)	3×10^{10} Hz
(C)	Microwaves	(III)	1×10^{21} Hz
(D)	X-rays	(IV)	1×10^{6} Hz

(1) A→(II); B→(I); C→(III); D→(IV)
(2) A→(I); B→(II); C→(IV); D→(III)
(3) A→(IV); B→(III); C→(II); D→(I)
(4) A→(II); B→(I); C→(IV); D→(III)

35. If M (A, Z) Mp and Mn denote the masses of the nucleus ${}^{A}_{Z}Z$, proton and neutron respectively in units of u and BE represents its binding energy in MeV, then
(1) $M(A, Z) = ZM_p + (A - Z)M_n + BE/c^2$
(2) $M(A, Z) = ZM_p + (A - Z)M_n - BE/c^2$
(3) $M(A, Z) = ZM_n + (A - Z)M_p + BE/c^2$
(4) $M(A, Z) = ZM_p + (A - Z)M_n + BE$

36. Given below are two statements: One is labelled as Assertion A and the other is labelled as Reason R:
Assertion (A): When a light wave travels from a rarer medium to a denser medium, the speed of the wave decreases, and the wavefront bends towards the normal.
Reason (R): According to Huygens' principle, the secondary wavelets in denser medium travel a smaller distance in the same time interval compared to the rarer medium.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true and R is NOT the correct explanation of A.

37. In YDSE two slits are placed 2 mm from each other, interference pattern is observed on a screen placed 1 m from the plane of slit. What is the fringe width for a light of wavelength 600 nm?
(1) 0.1 mm
(2) 0.2 mm
(3) 0.3 mm
(4) 0.4 mm

[Contd...

38. Two waves of equal frequencies have their amplitudes in the ratio of 4:3. They are superimposed on each other. The ratio of maximum and minimum intensities of the resultant wave is:

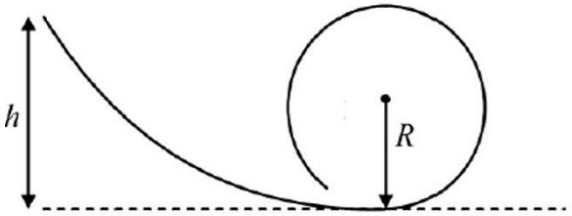
(1) 9:1

(2) 1:9

(3) 49:1

(4) 1:60

39. A body starts sliding down a frictionless track from a height $h = 5$ cm into a circular loop of radius R as shown in figure. For which among the following values of radius R, will the body fail to complete the vertical circle of the radius R is:



(1) 1 cm

(2) 2 cm

(3) 3 cm

(4) 0.5 cm

40. A train is speeding up with constant acceleration. Engine of train passes the single post speed 1 m/s and the middle coach (mid-point) of train passes the same signal post with speed 5 m/s. The speed of last compartment of train passes the post is:

(1) 7 m/s

(2) 6 m/s

(3) 8 m/s

(4) 9 m/s

41. A disc and a hollow sphere of the same mass have the same moments of inertia about their respective diameters, the ratio of their radii is:

(1) $(2)^{3/2}$: $(3)^{1/2}$

(2) $(3)^{1/2}$: $(2)^{1/2}$

(3) 3 : 2

(4) 2 : 3

42. Consider the following statements and select the **correct** statement(s).

I. Gravitational force may be attractive or repulsive

II. Gravitational force between two particles due to their masses is independent of presence of other particles.

III. Gravitational force is a short-range force

(1) I only

(2) II only

(3) II and III only

(4) I, II and III

43. An ideal gas undergoes a thermodynamic process where the pressure P varies with volume V according to the relation:

$P = P_0\left[1+\left(\frac{V}{V_0}\right)^2\right]$ where P_0 and V_0 are positive constants. The work done by the gas when its volume expands from $V = V_0$ to $V = 2V_0$ is:

(1) $\frac{10}{3}P_0V_0$

(2) $\frac{7}{3}P_0V_0$

(3) $\frac{4}{3}P_0V_0$

(4) $\frac{13}{3}P_0V_0$

44. A wave travelling in positive x-direction with amplitude A = 0.2m has a velocity of 360 m/s. If wavelength $\lambda = 60$m, then correct expression for the wave is:

(1) $y = 0.2\sin\left[2\pi\left(6t+\frac{x}{60}\right)\right]$m

(2) $y = 0.2\sin\left[\pi\left(6t+\frac{x}{60}\right)\right]$ m

(3) $y = 0.2\sin\left[2\pi\left(6t-\frac{x}{60}\right)\right]$ m

(4) $y = 0.2\sin\left[\pi\left(6t-\frac{x}{60}\right)\right]$ m

45. The core of transformer is laminated to reduce power leakage due to:

(1) flux leakage (2) hysteresis

(3) copper loss (4) eddy current

[Contd...

46. Highest lattice energy among the following is of:
(1) $MgCl_2$
(2) NaCl
(3) CaO
(4) Al_2O_3

47. Which among the following gases turn lime water milky?
(A) O_3
(B) SO_2
(C) NO_2
(1) B only
(2) A only
(3) B and C only
(4) A, B and C

48. The **correct** decreasing order of –I effect is:
(1) $-Br > -CHO > -\overset{+}{N}H_3$
(2) $-\overset{+}{N}H_3 > -Br > -CHO$
(3) $-\overset{+}{N}H_3 > -CHO > -Br$
(4) $-CHO > -Br > -\overset{+}{N}H_3$

49. What is the molality of aq. H_2SO_4 solution in which mole fraction of H_2SO_4 is 0.1?
(1) 5.23 m
(2) 4.81 m
(3) 7.45 m
(4) 6.17 m

50. Hybridisation of carbon in diamond and buckminsterfullerene respectively are:
(1) sp^2 and sp^3
(2) sp^2 and sp
(3) sp^3 and sp^2
(4) sp^2 and sp^2

51. The value of $E^{o}_{M^{2+}/M}$ will be:
Given: $E^{o}_{M^{2+}/M^{+}} = xV$ and $E^{o}_{M^{2+}/M} = yV$
(1) $(x + y)V$
(2) $\frac{1}{2}(x - y)V$
(3) $\frac{1}{2}(x + y)V$
(4) $2(x + y)V$

52. Given below are two statements:
Statement I: $[PtCl_4]^{2-}$ is square planar species.
Statement II: $[Ni(CN)_4]^{2-}$ is tetrahedral species.
In the light of the above statements, choose the *most appropriate* answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Statement I is correct but Statement II is incorrect.
(3) Statement I is incorrect but Statement II is correct.
(4) Both Statement I and Statement II are incorrect.

53. In which of the following equilibria, $K_p \neq K_c$?
(1) $2I(g) \rightleftharpoons I_2(g)$
(2) $N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$
(3) $2HI(g) \rightleftharpoons H_2(g) + I_2(g)$
(4) $CO(g) + H_2O(g) \rightleftharpoons H_2(g) + CO_2(g)$

54. Given below are two statements: one is labelled as Assertion A and other is labelled as Reason R:
Assertion A: Curdling of milk is an example of denaturation of proteins.
Reason R: Curdling of milk is caused due to the formation of lactic acid by the bacteria present in milk.
In the light of above statements, choose the **correct** answer from the options given below:
(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true and R is NOT the correct explanation of A.

55. Match **List-I** with **List-II**.

List-I (Element)		**List-II (Series)**	
(A)	Os	(I)	4d series
(B)	Ac	(II)	3d series
(C)	Sc	(III)	6d series
(D)	Ru	(IV)	5d series

Choose the **correct** answer from the options given below:
(1) A-III; B-I; C-IV; D-II
(2) A-IV; B-III; C-II; D-I
(3) A-II; B-IV; C-I; D-III
(4) A-I; B-III; C-IV; D-II

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56. For the redox reaction:

$MnO_4^- + C_2O_4^{2-} + H^+ \rightarrow Mn^{2+} + CO_2 + H_2O$

The **correct** coefficients of the reactants for the balanced reaction are:

	MnO_4^-	$C_2O_4^{2-}$	H^+
(1)	5	2	8
(2)	2	5	16
(3)	5	1	8
(4)	2	4	13

57. In an atom, the total number of electrons having quantum numbers $n = 4, |m_\ell| = 2$ and $m_s = +\frac{1}{2}$ is:

(1) 6
(2) 4
(3) 8
(4) 2

58. Which of the following binary mixture form maximum boiling azeotropes?

(1) $C_2H_5OH + H_2O$
(2) $C_6H_5OH + C_6H_5NH_2$
(3) $CH_3COCH_3 + C_2H_5OH$
(4) $CS_2 + CH_3COCH_3$

59. Given below are two statements:

Statement I: Melting point of n-pentane is less than that of neo-pentane.

Statement II: Boiling point of n-pentane is more than that of neo-pentane.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

(1) Both Statement I and Statement II are correct.
(2) Statement I is correct but Statement II is incorrect.
(3) Statement I is incorrect but Statement II is correct.
(4) Both Statement I and Statement II are incorrect.

60. The **correct** order of metallic character for the given elements is:

(1) Be < Mg < Na
(2) Mg < Be < Na
(3) Mg < Na < Be
(4) Na < Mg < Be

61. Given below are two statements: one is labelled as Assertion A and other is labelled as Reason R:

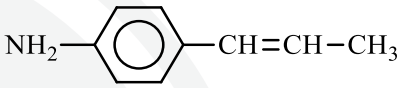
Assertion A: Aniline on nitration yields only ortho and para nitro derivatives of aniline.

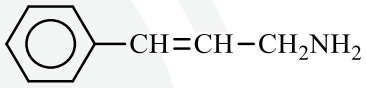
Reason R: The $-NH_2$ group in aniline is a powerful activating group.

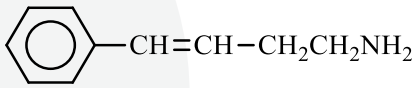
In the light of above statements, choose the **correct** answer from the options given below:

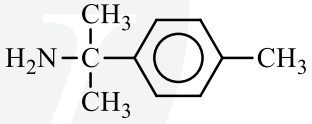
(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true and R is NOT the correct explanation of A.

62. In which of the following compounds, inductive, mesomeric and hyperconjugation effects are observed?

(1) 

(2) 

(3) 

(4) 

63. Given below are two statements:

Statement I: In solid state PCl_5 exist as an Ionic solid.

Statement II: The covalence of nitrogen in N_2O_5 is three.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

(1) Both Statement I and Statement II are correct.
(2) Statement I is correct but Statement II is incorrect.
(3) Statement I is incorrect but Statement II is correct.
(4) Both Statement I and Statement II are incorrect.

64. A reaction $X_2 + Y \rightarrow X_2Y$ take place as:

$X_2 \rightleftharpoons 2X$ (fast)

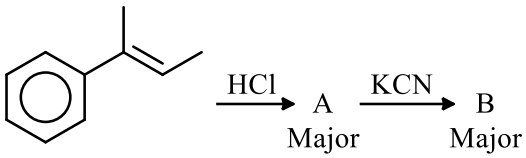
$2X + Y \rightarrow X_2Y$ (slow)

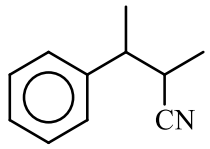
The order of the reaction is:

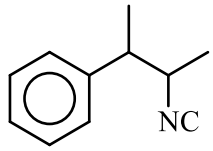
(1) 1
(2) 1.5
(3) 0.5
(4) 2

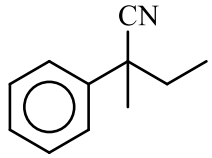
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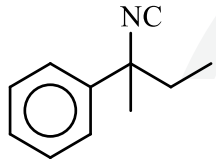
65. The product (B) formed in the following reaction sequence is:

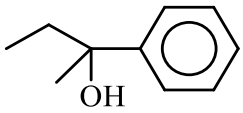


(1) 

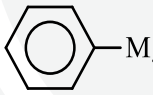
(2) 

(3) 

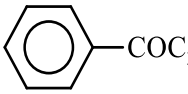
(4) 

66. 

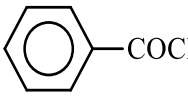
can be prepared by:

A. $CH_3CH_2COCH_3$ + 

followed by hydrolysis

B. 

+ CH_3MgBr followed by hydrolysis

C. 

+ C_2H_5MgBr followed by hydrolysis

(1) A and B only
(2) B and C only
(3) A and C only
(4) A, B and C

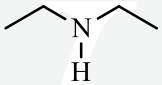
67. Among 3d series elements, enthalpy of atomisation is lowest and highest respectively for:
(1) Mn and Cr
(2) Zn and V
(3) Sc and Mn
(4) Zn and Cr

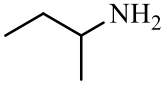
68. The dipole moment of KCl is 3.336×10^{-29} C m. If the interionic distance between K^+ and Cl^- is 2.6×10^{-10} m, the percentage ionic character of KCl is:
(1) 86.4% (2) 80.2%
(3) 74.1% (4) 92.6%

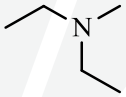
69. The **correct** statement among the following is/are:
(A) Silicon is very important component of glass
(B) Aluminium is most abundant metal
(C) Boron is low melting point non-metal
(1) A and B only
(2) B and C only
(3) A and C only
(4) A, B and C

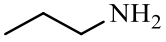
70. An ideal gas is allowed to expand from 10 L to 20 L against a constant external pressure of 2 atm. The work done is: (Given: 1 L-atm = 101.3 J)
(1) –1.01 kJ (2) –2.02 kJ
(3) –1.33 kJ (4) –2.84 kJ

71. An compound (A) on reaction with $PhSO_2Cl$ gives a compound which is insoluble in NaOH. The compound A is:

(1) 

(2) 

(3) 

(4) 

72. Match **List-I** with **List-II**.

	List-I (Reaction)		**List-II (Example)**
(A)	Metal displacement reaction	(I)	$2Na + 2H_2O \rightarrow 2NaOH + H_2$
(B)	Combination reaction	(II)	$NaCl + AgNO_3 \rightarrow AgCl + NaNO_3$
(C)	Non-redox reaction	(III)	$CuSO_4 + Zn \rightarrow Cu + ZnSO_4$
(D)	Non-metal displacement reaction	(IV)	$C + O_2 \rightarrow CO_2$

Choose the **correct** answer from the options given below:
(1) A-III; B-IV; C-II; D-I
(2) A-IV; B-III; C-II; D-I
(3) A-II; B-IV; C-I; D-III
(4) A-I; B-III; C-IV; D-II

[Contd...

73. The equilibrium, $A(g) + 6B(g) \rightleftharpoons 4C(g)$ is attained by mixing equal moles of A and B in an evacuated vessel. At equilibrium:

(1) [B] > [C]
(2) [B] > [A]
(3) [A] > [B]
(4) [C] > [A]

74. Match **List-I** with **List-II**.

List-I (Amino acid)		**List-II (Example)**	
(A)	Acidic amino acid	(I)	Glutamic acid
(B)	Basic amino acid	(II)	Glycine
(C)	Optically inactive amino acid	(III)	Cysteine
(D)	Sulphur containing amino acid	(IV)	Lysine

Choose the **correct** answer from the options given below:

(1) A-III; B-IV; C-II; D-I
(2) A-I; B-IV; C-II; D-III
(3) A-II; B-IV; C-I; D-III
(4) A-I; B-III; C-IV; D-II

75. Given below are two statements: one is labelled as Assertion A and other is labelled as Reason R:

Assertion A: An appropriate method for the preparation of t-butyl ether is:

$C_2H_5ONa + (CH_3)_3C-Cl \rightarrow (CH_3)_3C-OC_2H_5$

Reason R: C_2H_5ONa is a strong nucleophile.

In the light of above statements, choose the **correct** answer from the options given below:

(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true and R is NOT the correct explanation of A.

76. **Incorrect** statement regarding Henry's law is: (where: K_H is Henry's law constant)

(1) Different gases have different K_H values at same temperature.
(2) The value of K_H increases with increase of temperature.
(3) Higher the value of K_H at given pressure, higher the solubility of gas in liquids.
(4) Mole fraction of gas in the solution is proportional to the partial pressure of the gas over the solution.

77. Given below are two statements:

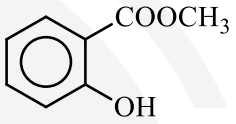
Statement I: Energies of the orbitals in the same subshell decrease with increase in the atomic number (Z_{eff}).

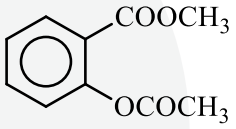
Statement II: For H-atom, energy of 3s and 3p orbitals are different.

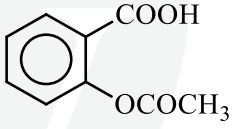
In the light of the above statements, choose the *most appropriate* answer from the options given below:

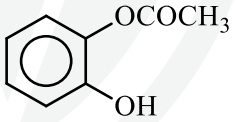
(1) Both Statement I and Statement II are correct.
(2) Statement I is correct but Statement II is incorrect.
(3) Statement I is incorrect but Statement II is correct.
(4) Both Statement I and Statement II are incorrect.

78. Aspirin is:

(1) 

(2) 

(3) 

(4) 

79. Match **List-I** with **List-II**.

List-I (Name reaction)		**List-II (Product obtainable)**	
(A)	Wurtz fittig reaction	(I)	Diphenyl
(B)	Fittig reaction	(II)	Chlorobenzene
(C)	Finkelstein reaction	(III)	Methylbenzene
(D)	Sandmeyer's reaction	(IV)	Ethyl iodide

Choose the **correct** answer from the options given below:

(1) A-III; B-I; C-IV; D-II
(2) A-I; B-IV; C-II; D-III
(3) A-II; B-IV; C-I; D-III
(4) A-I; B-III; C-IV; D-II

[Contd...

80. Match **List-I** with **List-II**.

List-I		List-II	
(A)	2.24 L of O_2 at STP	(I)	0.32 g
(B)	6.022×10^{21} molecules of O_2	(II)	32 g
(C)	1 g molecule of O_2	(III)	0.032 g
(D)	0.001 mole O_2	(IV)	3.2 g

Choose the **correct** answer from the options given below:

(1) A-III; B-I; C-IV; D-II
(2) A-IV; B-I; C-II; D-III
(3) A-II; B-IV; C-I; D-III
(4) A-I; B-III; C-IV; D-II

81. Hydration of but-1-yne in presence of $HgSO_4/H_2SO_4$ results the formation of:

(1) Butanal
(2) Butan-2-one
(3) Butan-1-ol
(4) Butan-2-ol

82. The **correct** order of boiling point of given hydrides is:

(1) $SbH_3 > NH_3 > PH_3$
(2) $NH_3 > SbH_3 > PH_3$
(3) $NH_3 > PH_3 > SbH_3$
(4) $SbH_3 > PH_3 > NH_3$

83. The **correct** decreasing order of pH of 0.1M solutions of given electrolytes is:

(1) $HCl > NH_4Cl > CH_3COONa > NaCl$
(2) $NH_4Cl > CH_3COONa > HCl > NaCl$
(3) $NaCl > HCl > NH_4Cl > CH_3COONa$
(4) $CH_3COONa > NaCl > NH_4Cl > HCl$

84. Given below are two statements:

Statement I: Brown ring test is used for the conformation of Cl^- ion.

Statement II: The colour of lead sulphide precipitate is pink.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

(1) Both Statement I and Statement II are correct.
(2) Statement I is correct but Statement II is incorrect.
(3) Statement I is incorrect but Statement II is correct.
(4) Both Statement I and Statement II are incorrect.

85. If the conductivity of 0.001 M HA is 4×10^{-5} S cm^{-1}, then degree of dissociation of HA will be:

(Given: Λ_m^0 of HA = 400 S cm^2 mol^{-1})

(1) 0.05
(2) 0.15
(3) 0.25
(4) 0.10

86. In the reaction: $SO_2(g) + \frac{1}{2}O_2(g) \rightarrow SO_3(g)$, if at a certain instant of time the rate of disappearance of $SO_2(g)$ is 32 g s^{-1} then the rate of appearance of $SO_3(g)$ at the same instant will be:

(1) 64 g s^{-1}
(2) 100 g s^{-1}
(3) 40 g s^{-1}
(4) 32 g s^{-1}

87. The number of possible enantiomeric pairs that can be obtained by monochlorination of 2,3-dimethylbutane is:

(1) 0
(2) 1
(3) 2
(4) 3

88. Which of the following complexes will exhibit maximum molar conductance?

(1) $CoCl_3 \cdot 3NH_3$
(2) $CoCl_3 \cdot 4NH_3$
(3) $CoCl_3 \cdot 5NH_3$
(4) $CoCl_3 \cdot 6NH_3$

89. The **correct** relation for 2 moles of an ideal gas is: (where, C_p and C_v are molar heat capacity at constant pressure and constant volume respectively)

(1) $C_p = R + C_v$
(2) $C_p = 2R - C_v$
(3) $2C_p = R - 2C_v$
(4) $C_p = R + \frac{C_v}{2}$

90. Among Mg, Mg^{2+}, Al and Al^{3+}, which of the following species will have the largest and smallest size respectively?

(1) Al and Mg^{2+}
(2) Mg and Al^{3+}
(3) Mg and Mg^{2+}
(4) Al and Al^{3+}

[Contd...

91. How many of the following statements regarding a prokaryotic cell are correct?

A. The cytoplasm is the main arena of cellular activities.
B. Ribosomes are the only organelles found.
C. The function of eukaryotic mitochondria is performed by chromatophores.
D. The organisation of the prokaryotic cell is fundamentally similar despite a wide variety of shapes and functions.
E. Mycoplasmas are the smallest cells.

Choose the **correct** option.

(1) Five (2) Four only
(3) Three only (4) Two only

92. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Seeds within fruits	(I)	*Salvinia*
(B)	Seeds are naked	(II)	*Sphagnum*
(C)	Event precursor to seed habit	(III)	*Eucalyptus*
(D)	Main plant body is gametophyte	(IV)	*Sequoia*

Choose the *most appropriate* answer from the options given below.

(1) A-II, B-III, C-I, D-IV
(2) A-IV, B-II, C-III, D-I
(3) A-I, B-III, C-II, D-IV
(4) A-III, B-IV, C-I, D-II

93. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): In bacteria, transcription and translation can be coupled.

Reason (R): The mRNA in bacteria does not contain introns and is non-functional.

In the light of the above statements, choose the *most appropriate* answer from the options given below.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(3) (A) is true but (R) is false.
(4) (A) is false but (R) is true.

94. The cells are barrel-shaped in;

(1) pericycle of dicot stem.
(2) endodermis of dicot root.
(3) hypodermis of dicot stem.
(4) palisade parenchyma of dorsiventral leaf.

95. Identify the **incorrect** statement about 'reductionist biology' approach.

(1) It is a physico-chemical approach to study and understand living organisms.
(2) This approach enables us to describe the various physiological and behavioural processes in molecular terms.
(3) The approach is established by analysis of living tissues for elements and compounds.
(4) It cannot explain the abnormal processes that occur during any diseased condition.

96. Arrange the following structures in the **correct** sequence of their formation from the earliest to the last in a typical angiosperm.

A. Megaspore B. Embryo
C. Endosperm D. Embryo sac
E. Fruit

Choose the **correct** option.

(1) A → D → C → B → E
(2) B → C → E → A → D
(3) B → C → A → E → D
(4) C → B → A → E → D

97. Given below are two statements:

Statement I: During chemiosmosis, the primary electron acceptor donates the electrons to FNR.

Statement II: Chemiosmosis leads to reduction of pH in thylakoid lumen.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

98. A pea plant heterozygous for plant height and seed shape is subjected to a test cross. What proportion of the progeny will be phenotypically identical to the recessive parent?

(1) 1/2 (2) 1/4
(3) 1/8 (4) 3/4

99. Identify the **incorrectly** matched pair.

(1) G_1 phase - Cell is metabolically active and continuously grows
(2) S phase - Centriole duplication in animal cells
(3) G_0 phase - Cells are not metabolically active
(4) M phase - Phase when the actual cell division or mitosis occurs

[Contd...

100. How many of the organisms listed below can be considered as primary consumers in their respective ecosystems?

A. Insects
B. Birds
C. Mammals
D. Molluscs
E. Zooplankton

Choose the **correct** option.

(1) Two only (2) Three only
(3) Four only (4) Five

101. Match the **List-I** with **List-II**.

List-I		List-II	
(A)	*Trypanosoma*	(I)	Ciliated and move actively
(B)	*Paramoecium*	(II)	Produces spores and pathogenic to humans
(C)	*Plasmodium*	(III)	Possess pseudopodia and is parasitic
(D)	*Entamoeba*	(IV)	Flagellated and pathogenic to humans

Choose the *most appropriate* answer from the options given below:

(1) A-IV, B-I, C-II, D-III
(2) A-IV, B-II, C-I, D-III
(3) A-I, B-IV, C-II, D-III
(4) A-III, B-I, C-IV, D-II

102. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): Species diversity generally decreases as we move away from the equator towards the poles.

Reason (R): In a forest of equal area, the number of plant species in New York is more than that in Equador.

In the light of the above statements, choose the *most appropriate* answer from the options given below.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(3) (A) is true but (R) is false.
(4) (A) is false but (R) is true.

103. A monocot plant family among the following is;

(1) Compositae.
(2) Poaceae.
(3) Cruciferae.
(4) Leguminosae.

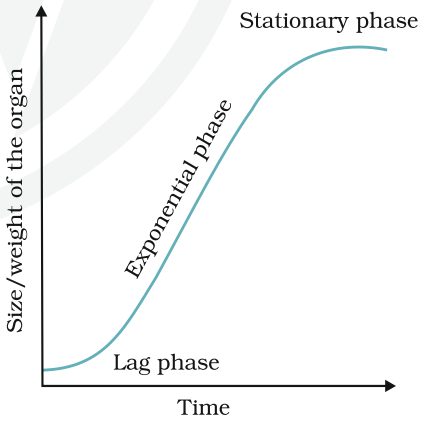
104. Read the following statements.

A. Lactic Acid Bacteria improve the nutritional quality of milk by increasing vitamin B_2.
B. The dough used for making *dosa* is fermented by fungi.
C. 'Toddy' is a traditional drink made by fermenting sap from palms.
D. The large holes in 'Swiss cheese' are due to the production of a large amount of CO_2 by *Propionibacterium sharmanii*.
E. *Saccharomyces cerevisiae* is used for fermenting malted cereals and fruit juices to produce ethanol.

In the light of above statements choose the **correct** answer from the options given below:

(1) B and C are incorrect and A, D and E are correct
(2) D and E are incorrect and A, B and C are correct
(3) C and D are incorrect and A, B and E are correct
(4) A and B are incorrect and C, D and E are correct

105. Given below is an idealised growth curve typical of a plant organ. Choose the **correct** statement about it.



(1) It can be mathematically expressed as $L_t = L_0 + rt$.
(2) Such a curve is shown by a root elongating at a constant rate.
(3) A living organism growing in a natural environment usually shows such growth pattern.
(4) In such a growth, following mitotic cell division, only one daughter cell continues to divide while the other differentiates and matures.

[Contd...

106. Read the following statements about phenylketonuria.
A. It is an inborn error of metabolism.
B. It is inherited as an autosomal dominant trait.
C. The affected individual lacks an enzyme that converts phenylalanine into tyrosine.
D. It is characterized by mental retardation and a reduction in hair and skin pigmentation.
E. It is caused by a mutation in more than three genes.
The **correct** statements are;
(1) A, C and D only
(2) B, D and E only
(3) A, B, C and D only
(4) A, B, C, D and E

107. Given below are two statements:
Statement I: Fermentation accounts for only a partial breakdown of glucose.
Statement II: In fermentation, there is a net gain of only two molecules of ATP for each molecule of glucose degraded to pyruvic acid.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

108. A dicotyledonous stem;
(1) has closed vascular bundles.
(2) has endarch primary xylem.
(3) has radially placed xylem and phloem.
(4) usually has xylem present on the outside of phloem.

109. Pseudocopulation is shown by the pollinator of;
(1) *Yucca*. (2) *Amorphophallus.*
(3) *Ophrys.* (4) fig.

110. The housefly is classified in all the following taxonomical categories, **except**;
(1) phylum Arthropoda.
(2) order Diptera.
(3) class Insecta.
(4) genus *Drosophila*.

111. Match the **List-I** with **List-II**.

List-I		List-II	
(A)	Catalytic RNA	(I)	tRNA
(B)	Soluble RNA	(II)	Primary transcript in eukaryotes
(C)	hnRNA	(III)	Formed after splicing in eukaryotes
(D)	mRNA	(IV)	Ribozyme

Choose the *most appropriate* answer from the options given below.
(1) A-I, B-II, C-III, D-IV
(2) A-IV, B-I, C-II, D-III
(3) A-II, B-I, C-IV, D-III
(4) A-IV, B-III, C-II, D-I

112. The substance peat is obtained from species of;
(1) *Marchantia*. (2) *Sphagnum*.
(3) *Polytrichum*. (4) *Adiantum*.

113. Different species occupying different levels in a forest ecosystem were vertically distributed. Choose the **correct** statement about it.
(1) This is called stratification.
(2) It is a functional unit of the forest ecosystem.
(3) The top strata is occupied by grasses.
(4) The bottom strata is occupied by shrubs.

114. Identify the **incorrect** statement about phosphoenol pyruvate.
(1) When it is converted to pyruvate in cytoplasm, ATP is synthesised by substrate-level phosphorylation.
(2) It is a 3-carbon compound.
(3) It is the reducing agent of CO_2 in bundle sheath cells of C_4 plants.
(4) The enzyme PEPcase catalyses its conversion to oxaloacetate.

115. Identify the **correctly** matched pair.
(1) Golgi apparatus – Double-membrane bound organelle
(2) Rough ER – Bears ribosomes on surface
(3) Lysosomes – Formed by packaging in the ER
(4) Mitochondria – Site of anaerobic respiration

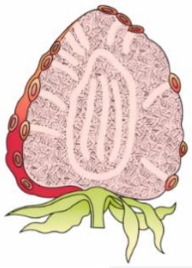
116. Given below are two statements:
Statement I: In a DNA helix, Adenine and Thymine are paired by 3 hydrogen bonds.
Statement II: In a DNA helix, the backbone is composed of sugar and nitrogenous base.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both Statement I and Statement II are incorrect.
(2) Both Statement I and Statement II are correct.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

[Contd...

117. At how many steps in glycolysis is ATP utilised in one turn?
(1) 4
(2) 10
(3) 3
(4) 2

118. Which of the following is an **incorrect** statement about a flower?
(1) It is a modified shoot meant for sexual reproduction.
(2) When a flower has both androecium and gynoecium, it is bisexual.
(3) Flowers with reduced leaves at the base of pedicel are called ebracteate.
(4) The arrangement of flowers on the floral axis is termed as inflorescence.

119. The fruit shown in the given diagram;



(1) develops parthenogenetically.
(2) is formed by contribution from thalamus apart from ovary.
(3) is a product of apomixis.
(4) is a true fruit.

120. The bulliform cells are present on the;
(1) adaxial surface of dorsiventral leaf.
(2) abaxial surface of dorsiventral leaf.
(3) adaxial surface of grass leaf.
(4) abaxial surface of grass leaf.

121. A student observed a filamentous organism that had two types of cells. One type of cells performed photosynthesis, and the other one specialized for nitrogen fixation. Choose the correct statement about it.
(1) The cell that performs photosynthesis is called heterocyst.
(2) This organism is considered an archaebacterium.
(3) Colonies of such organisms are generally surrounded by gelatinous sheath.
(4) It is a chemosynthetic autotroph.

122. Match the **List-I** with **List-II**.

List-I		List-II	
(A)	*Glomus*	(I)	Effective biocontrol agent of several plant pathogens and source of cyclosporin A
(B)	*Nucleopolyhedrovirus*	(II)	Members form mycorrhizal associations
(C)	*B. thuringiensis*	(III)	Acellular organism used as biocontrol agent
(D)	*Trichoderma*	(IV)	Biocontrol agent of butterfly caterpillars

Choose the *most appropriate* answer from the options given below.
(1) A-II, B-III, C-IV, D-I
(2) A-III, B-II, C-I, D-IV
(3) A-II, B-I, C-IV, D-III
(4) A-IV, B-III, C-II, D-I

123. In pteridophytes, the main plant body is:
(1) haploid and thallus-like.
(2) differentiated into true root, stem and leaves.
(3) devoid of well-differentiated vascular tissues.
(4) attached to the photosynthetic gametophyte.

124. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A): The number of autosomes in a normal and in an individual with Klinefelter's syndrome is same.
Reason (R): In an individual with Klinefelter's syndrome, only one X chromosome is present.
In the light of the above statements, choose the *most appropriate* answer from the options given below.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(3) (A) is true but (R) is false.
(4) (A) is false but (R) is true.

125. How many of the following are predominantly plant growth promoters?
A. Auxins
B. Gibberellins
C. Cytokinins
D. Abscisic acid
E. Ethylene
Choose the **correct** option.
(1) Two only (2) Three only
(3) Four only (4) Five

[Contd...

126. The human genome consists of;
(1) 98% DNA that codes for proteins.
(2) about 3,00,000 genes.
(3) a very small portion of repeated sequences.
(4) 3164.7 million bp.

127. Identify the **incorrect** statement.
(1) Mitochondria are not considered part of the endomembrane system.
(2) The Golgi cisternae are concentrically arranged near the nucleus with distinct convex *cis* and concave *trans* faces.
(3) The nucleus was discovered by Antonie Von Leeuwenhoek.
(4) Microbodies contain enzymes.

128. All the following ecological pyramids are upright, **except**
(1) Pyramid of numbers in a tree ecosystem.
(2) Pyramid of biomass in a forest ecosystem.
(3) Pyramid of energy in grassland ecosystem.
(4) Pyramid of numbers in a grassland ecosystem.

129. The accessory photosynthetic pigments in higher plants;
(1) form the reaction centre of the photosystems.
(2) prevent photo-oxidation of the reaction centre.
(3) include only chlorophyll *b*, chlorophyll *a* and xanthophylls.
(4) are present in stroma of the chloroplasts.

130. Which of the following was **not** a limitation of the two-kingdom classification system?
(1) No distinction between eukaryotes and prokaryotes.
(2) No distinction between unicellular and multicellular organisms.
(3) No distinction between photosynthetic and non-photosynthetic organisms.
(4) No distinction between plants and animals.

131. In the cell cycle of a drone (male honey bee), which of the following events will not be observed?
(1) Duplication of DNA
(2) Crossing over
(3) Splitting of centromere
(4) Cytokinesis by formation of cell furrow

132. Given below are two statements:
Statement I: In mutualism, both interacting species benefit, while in competition, both are harmed.
Statement II: In amensalism, one species is harmed and the other is benefited.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

133. Underground stems of which of the following sets of plants are modified to store food and also act as organs of perennation to tide over conditions unfavourable for growth?
(1) Cucumber, pumpkins, watermelon
(2) *Citrus*, *Bougainvillea*
(3) Potato, ginger, turmeric
(4) Grass, strawberry

134. Match the **List-I** with **List-II**.

List-I (Trait being inherited)		**List-II (Characteristic)**	
(A)	Flower colour in dogflower	(I)	Polygenic inheritance
(B)	Flower colour in pea	(II)	Incomplete dominance
(C)	Skin colour in humans	(III)	Complete dominance
(D)	ABO blood grouping in humans	(IV)	Multiple alleles

Choose the *most appropriate* answer from the options given below.
(1) A-II, B-III, C-I, D-IV
(2) A-IV, B-II, C-III, D-I
(3) A-II, B-I, C-III, D-IV
(4) A-III, B-IV, C-I, D-II

135. Which of the following devices in flowering plants prevent autogamy but not geitonogamy?
A. Pollen release and stigma receptivity are not synchronised.
B. Anther and stigma are placed at different positions.
C. Production of unisexual flowers on the same plant (monoecious).
D. Self-incompatibility.
E. Production of flowers on different plants (dioecy).
Choose the **correct** option.
(1) B, C and D (2) C, D and E
(3) A, B and C (4) A, D and E

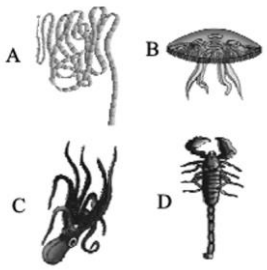
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136. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: Platyhelminthes are considered to have an incomplete digestive system.
Reason R: They possess a body cavity that is not completely lined by mesoderm.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both A and R are true and R is the correct explanation of A.
(2) Both A and R are true but R is not the correct explanation of A.
(3) A is true but R is false.
(4) Both A and R are false.

137. Consider the following statements regarding biomolecules.
A. Collagen is the most abundant protein in the whole of the biosphere.
B. Lysozyme are nucleic acids with catalytic power.
C. A competitive inhibitor binds to the active site of the enzyme and inhibits the activity of the enzyme.
D. Arachidonic acid has 20 carbon atoms including the carboxyl carbon.
E. Cellulose is a homopolymer consisting of only fructose as its building blocks.
How many of the above statements are **correct**?
(1) Two (2) Three
(3) Four (4) Five

138. In *Periplaneta americana*, the Malpighian tubules are situated at the junction of:
(1) Pharynx and oesophagus
(2) Foregut and midgut
(3) Midgut and hindgut
(4) Ileum and colon

139. The figure shows four animals A, B, C and D. Select the **correct** answer with respect to common characteristics of two of these animals.



(1) A and D have cnidoblasts for self-defense
(2) C and D have a true coelom
(3) A and D respire mainly through bodywall
(4) B and C show radial symmetry

140. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Lub	(I)	Secrete histamine, serotonin, and heparin
(B)	Basophils	(II)	First heart sound
(C)	Dub	(III)	B and T forms responsible for immune responses
(D)	Lymphocytes	(IV)	Associated with the closure of the semilunar valves

Choose the **correct** answer from the options given below:
(1) A-I, B-II, C-IV, D-III
(2) A-II, B-IV, C-I, D-III
(3) A-II, B-I, C-IV, D-III
(4) A-III, B-I, C-IV, D-II

141. If the stroke volume of a healthy adult human is 70 mL and the heart rate is 72 beats per minute, what is the approximate volume of blood filtered by the kidneys per minute?
(1) 5000 mL (2) 125 mL
(3) 1100 mL (4) 180 Litres

142. Select the **correct** statements regarding the sliding filament theory.
A. A motor neuron along with the muscle fibres connected to it constitutes a motor unit.
B. A neural signal reaching the neuromuscular junction releases the neurotransmitter adrenaline.
C. Calcium ions bind with a subunit of troponin on actin filaments to remove the masking of active sites for myosin.
D. During the shortening of the muscle, the A bands get reduced while the I bands retain their length.
E. The myosin head acts as an active ATPase enzyme.
(1) A, B and C (2) A, C and E
(3) B, D and E (4) C, D and E

143. The inner parts of the cerebral hemispheres and a group of associated deep structures like the amygdala and hippocampus form a complex structure called the:
(1) Cerebellar cortex
(2) Limbic system
(3) Brain stem
(4) Corpora quadrigemina

[Contd...

144. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Erythropoietin	(I)	Atrial wall of heart
(B)	Gastrin	(II)	Juxtaglomerular cells of kidney
(C)	Secretin	(III)	Stimulates secretion of HCl and pepsinogen
(D)	Atrial natriuretic factor	(IV)	Stimulates secretion of water and bicarbonate ions

Choose the **correct** answer from the options given below:

(1) A-II, B-III, C-IV, D-I
(2) A-I, B-IV, C-III, D-II
(3) A-II, B-IV, C-III, D-I
(4) A-I, B-III, C-IV, D-II

145. Given below are two statements:

Statement I: Rapid secretion of LH leading to its maximum level during the mid-cycle induces rupture of the Graafian follicle.

Statement II: Uterine fundus is not a part fallopian tube.

In the light of above statements, choose the most appropriate answer from the options given below:

(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

146. 'Saheli', a non-steroidal oral contraceptive for females was developed by scientists at:

(1) ICMR, New Delhi
(2) CDRI, Lucknow
(3) IARI, New Delhi
(4) NEERI, Nagpur

147. In a randomly mating population at genetic equilibrium, the frequency of a recessive allele is 0.4. What will be the percentage of individuals carrying the dominant phenotype in this population?

(1) 16%
(2) 36%
(3) 84%
(4) 48%

148. How many of the following pairs of diseases and their related features are **incorrectly** matched?

I. Typhoid - *Salmonella typhi* - Confirmed by Widal test
II. Pneumonia - *Haemophilus influenzae* - Alveoli get filled with fluid
III. Malaria - *Plasmodium falciparum* - Malignant and most serious malaria
IV. Elephantiasis - *Wuchereria bancrofti* - Slowly developing chronic inflammation of lymphatic vessels of upper limbs usually

(1) One (2) Two
(3) Three (4) Four

149. Which of the following represents the **correct** sequence of steps in a PCR?

(1) Extension, Denaturation, Annealing
(2) Denaturation, Annealing, Extension
(3) Annealing, Extension, Denaturation
(4) Denaturation, Extension, Annealing

150. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: RNA interference (RNAi) takes place in all eukaryotic organisms as a method of cellular defense.

Reason R: It involves the silencing of a specific mRNA due to a complementary double-stranded RNA (dsRNA) molecule that binds to and prevents translation of the mRNA.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) Both A and R are true and R is the correct explanation of A.
(2) Both A and R are true but R is not the correct explanation of A.
(3) A is true but R is false.
(4) Both A and R are false.

151. Which of the following phyla consists of animals that are bilaterally symmetrical, triploblastic, and pseudocoelomate?

(1) Platyhelminthes
(2) Aschelminthes
(3) Annelida
(4) Arthropoda

152. Which of the following nitrogenous bases is a substituted pyrimidine exclusively found in DNA molecules?

(1) Adenine (2) Uracil
(3) Thymine (4) Cytosine

[Contd...

153. Read the following statements regarding animal tissues carefully.

A. Compound epithelium consists of two or more cell layers and plays a major role in secretion and absorption.

B. The cuboidal epithelium is commonly found in ducts of glands and tubular parts of nephrons in kidneys.

C. Ciliated epithelium is mainly present in the inner surface of hollow organs like bronchioles and fallopian tubes.

D. Tight junctions facilitate the cells to communicate with each other by connecting the cytoplasm of adjoining cells.

E. Gap junctions help to stop substances from leaking across a tissue.

Choose the **correct** option:

(1) A and E are incorrect
(2) A, D and E are correct
(3) B, C and D are incorrect
(4) B and C are correct

154. The volume of air remaining in the lungs even after a forcible expiration is known as:

(1) TV
(2) ERV
(3) RV
(4) FRC

155. In a standard electrocardiogram (ECG) of a normal person, the T-wave represents the:

(1) Electrical excitation of the atria
(2) Depolarisation of the ventricles
(3) Return of the ventricles from excited to normal state
(4) Initiation of ventricular systole

156. The counter-current mechanism in the human kidney is mainly responsible for:

(1) Increasing blood pressure
(2) Maintaining medullary osmotic gradient for urine concentration
(3) Stimulating glomerular filtration rate only
(4) Increasing cardiac output

157. Which pairs of ribs in humans are false ribs?

(1) 8th, 9th and 10th pairs
(2) 11th and 12th pairs
(3) 1st to 7th pairs
(4) 9th, 10th and 11th pairs

158. Given below are two statements:

Statement I: Impulse transmission across an chemical synapse is faster than that across a electrical synapse.

Statement II: Diencephalon is the part of forebrain.

In the light of above statements, choose the most appropriate answer from the options given below:

(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

159. Deficiency of iodine is directly associated with:

(1) Hyperthyroidism
(2) Hypothyroidism
(3) Exopthalmic goitre
(4) Graves' disease

160. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: The secondary oocyte formed during oogenesis retains the bulk of the nutrient-rich cytoplasm of the primary oocyte.

Reason R: The first meiotic division of the primary oocyte is an unequal division resulting in a large haploid secondary oocyte and a tiny first polar body.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) Both A and R are true and R is the correct explanation of A.
(2) Both A and R are true but R is not the correct explanation of A.
(3) A is true but R is false.
(4) Both A and R are false.

161. Which of the following is an example of a hormone-releasing Intra Uterine Device (IUD)?

(1) Lippes loop
(2) Multiload 375
(3) LNG-20
(4) CuT

[Contd...

162. Given below are two statements:

Statement I: The process of evolution of different species in a given geographical area starting from a point and literally radiating to other areas of geography is called adaptive radiation.

Statement II: When more than one adaptive radiation appeared to have occurred in an isolated geographical area, it is referred to as divergent evolution.

In the light of above statements, choose the most appropriate answer from the options given below:

(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

163. Interferons are component of which type of innate immunity barrier?

(1) Physical barriers
(2) Cellular barriers
(3) Physiological barriers
(4) Cytokine barriers

164. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Lysozyme	(I)	Breaks down fungal cell wall
(B	Cellulase	(II)	Joins DNA fragments
(C)	Chitinase	(III)	Breaks down bacterial cell wall
(D)	DNA ligase	(IV)	Breaks down plant cell wall

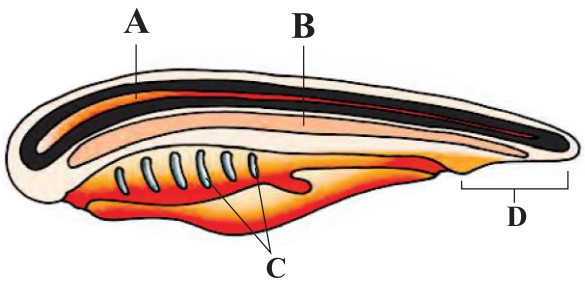
Choose the **correct** answer from the options given below:

(1) A-IV, B-III, C-I, D-II
(2) A-III, B-IV, C-I, D-II
(3) A-III, B-I, C-IV, D-II
(4) A-I, B-IV, C-III, D-II

165. In genetically modified Bt cotton, the inactive protoxin produced by *Bacillus thuringiensis* is converted into an active form of toxin in the insect gut due to:

(1) Acidic pH of the gut which denatures the protein
(2) Alkaline pH of the gut which solubilises the crystals
(3) Action of specific proteases in the midgut
(4) Low temperature in the insect gut

166. 'A' points to the dorsal hollow tubular structure, 'B' points to the solid rod-like structure lying just below 'A', and 'C' points to the paired openings in the pharynx.



Identify the **correct** structures for A and B respectively w.r.t chordates.

(1) A - Notochord, B - Nerve cord
(2) A - Nerve cord, B - Notochord
(3) A - Gill slits, B - Post-anal tail
(4) A - Nerve cord, B - Gill slits

167. Which of the following organic compounds is classified as a primary metabolite?

(1) Anthocyanins
(2) Morphine
(3) Amino acids
(4) Lemon grass oil

168. In the anatomy of a frog, the special venous connection existing between the liver and the intestine is called the:

(1) Renal portal system
(2) Systemic circulation
(3) Pulmonary circulation
(4) Hepatic portal system

169. In the human body, the highest percentage (about 70%) of carbon dioxide is transported from tissues to the lungs:

(1) As carbamino-haemoglobin
(2) In a dissolved state through plasma
(3) As bicarbonate ions in the plasma
(4) As carboxyhaemoglobin

170. Identify the **incorrect** pair regarding human blood groups and their corresponding antibodies naturally present in the plasma:

(1) Blood Group A : anti-B
(2) Blood Group B : anti-A
(3) Blood Group AB : anti-A, anti-B
(4) Blood Group O : anti-A, anti-B

[Contd...

171. Diuresis is prevented by:
(1) Renin from JG cells via switching off the osmoreceptors
(2) ANF from atria of the heart
(3) Aldosterone from adrenal medulla
(4) Vasopressin from Neurohypophysis

172. Evaluate the following statements regarding human skeletal joints and select the **correct** sequence of True (T) or False (F).
A. Fibrous joints allow considerable movement and are vital for locomotion.
B. In cartilaginous joints, the involved bones are joined together with the help of cartilages.
C. Synovial joints are characterized by the presence of a fluid-filled synovial cavity.
D. The joint between the atlas and axis vertebrae is an example of a hinge joint.
(1) F, T, T, F (2) T, F, T, F
(3) F, T, F, T (4) F, F, T, T

173. At resting condition, the axonal membrane is comparatively more permeable to:
(1) Na^+ (2) K^+
(3) Ca^{2+} (4) Both Na^+ and K^+

174. Which of the following is an example of such a hormone that interacts with intracellular receptors?
(1) Glucagon (2) Epinephrine
(3) Estradiol (4) Insulin

175. Match **List-I List-II**.

List-I		List-II	
(A)	Leydig cells	(I)	Provide nutrition to the male germ cells
(B)	Sertoli cells	(II)	Synthesise and secrete androgens
(C)	Bulbourethral glands	(III)	Site of storage and maturation of sperms
(D)	Epididymis	(IV)	Help in the lubrication of penis

Choose the **correct** answer from the options given below:
(1) A-II, B-I, C-IV, D-III
(2) A-I, B-II, C-III, D-IV
(3) A-II, B-IV, C-I, D-III
(4) A-III, B-I, C-IV, D-II

176. According to the Medical Termination of Pregnancy (Amendment) Act, 2017, MTPs are considered relatively safe up to which period of pregnancy?
(1) First trimester (2) Up to 20 weeks
(3) Up to 24 weeks (4) Up to 36 weeks

177. The flippers of Penguins and Dolphins is a classic example of:
(1) Divergent evolution
(2) Analogous organs
(3) Adaptive radiation
(4) Homologous organs

178. Given below are two statements:
Statement I: HIV is a malady that can only be tackled, by the society and medical fraternity acting together, to spread the disease.
Statement II: The infected macrophages continue to produce virus particles and thus act like an HIV factory.
In the light of above statements, choose the most appropriate answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

179. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: *Eco*RI leaves sticky ends on the DNA strands.
Reason R: It cuts the DNA strand a little away from the centre of the palindrome sites.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both A and R are true and R is the correct explanation of A.
(2) Both A and R are true but R is not the correct explanation of A.
(3) A is true but R is false.
(4) Both A and R are false.

180. Why was first clinical gene therapy **not** a permanent cure to ADA deficiency?
(1) The introduced normal ADA gene mutates rapidly in the blood.
(2) The genetically engineered lymphocytes are not immortal and have a definite life span.
(3) The retroviral vector used inherently destroys the host immune cells over time.
(4) The ADA enzyme is rapidly degraded by proteases present in the serum.

■■■

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